ASSIGNMENT NO. 2

BIO203: METHODS IN MOLECULAR BIOLOGY

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Inbox for LMS activities on Paid Basis (03064963933)

RT-PCR vs. Multiplex PCR

Factor	Reverse Transcriptase PCR	Multiplex PCR
Template Used	RNA	DNA
Number of Reactions	Single reaction per target	Single reaction for multiple targets
Number of Primers	Two primers for each target	Multiple pairs of primers in the same reaction (each pair for a different target)

Factor	Reverse Transcriptase PCR	Multiplex PCR
Enzyme Used	Reverse Transcriptase and Taq polymerase	Taq polymerase

Please note:

1. **Template Used:** In RT-PCR, the starting material is RNA, which is reverse-transcribed into complementary DNA (cDNA) using the enzyme reverse transcriptase. In multiplex PCR, the starting material is usually DNA.
2. **Number of Reactions:** In RT-PCR, there is usually a single reaction for each target of interest. However, in multiplex PCR, multiple targets are amplified in a single reaction.
3. **Number of Primers:** In RT-PCR, each target requires one pair of primers. For multiplex PCR, there are several pairs of primers in the same reaction, each pair specific for a different target.
4. **Enzyme Used:** RT-PCR uses reverse transcriptase to create cDNA from RNA, and then Taq polymerase is used for amplification of the DNA. Multiplex PCR only uses Taq polymerase for amplifying the DNA, as the starting material is DNA.